Software Requirements Specification (SRS)

AI Prompt-based Circuit Designer

Institute: Zeal College of Engineering & Research, Pune-41 (An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Department: Electronics and Computer Engineering **Domain:** Artificial Intelligence / Machine Learning, Web Application, Electronic Circuit Design

Member	Name	Roll No.	ZPRN
1	Patare Anushree Ramesh	EC2243	125UEC1099
2	Ambhore Praful Raju	EC2252	125UEC1160
3	Sontakke Sanket Dhanraj	EC2258	125UEC1066
4	Kachare Sejal Santosh	EC2266	125UEC1174

1. Introduction

1.1 Purpose

The purpose of the AI Prompt-Based Circuit Designer is to provide an AI-assisted system that allows users to describe their desired electronic circuit using simple natural-language prompts (e.g., "Design a circuit to control an LED using a push button"). The system will automatically interpret these prompts, identify required components and connections, and generate a corresponding visual circuit diagram, making circuit design more accessible, faster, and easier for beginners.

1.2 Problem Statement

Traditional circuit-design systems require users to have prior knowledge of electronic components, symbols, connections, and complex software tools, which poses a significant barrier for beginners and students. The proposed system addresses this by developing a user-friendly prompt-based designer that understands natural-language requirements and automatically generates a structured circuit diagram, thereby reducing manual effort and the initial learning curve.

2. Overall Description

2.1 Users

Students / Beginners: Users who need to quickly create preliminary circuit designs without manually drawing every connection.

Educators: Faculty who can use the tool to provide an educational understanding of components and their connections to students.

2.2 System Modules

Prompt Processing (NLP) Module: Analyzes the natural language circuit prompt from the user.

Component & Requirement Identification Module: Extracts components, values, input/output requirements, and relationships.

Circuit Design Logic Module: Maps extracted requirements to a predefined component library and connection rules/topologies.

Circuit Generation Module: Creates the required circuit connections and renders the schematic.

Output & Explanation Module: Displays the visual circuit diagram, component list, and basic educational explanations.

3. Functional Requirements

FR-1: Natural Language Interface

The system shall provide a simple and beginner-friendly user interface to accept natural-language descriptions of the required circuit.

FR-2: Prompt Processing & AI Interpretation

The system shall utilize Natural Language Processing (NLP) and Large Language Models (LLM) to process the input prompt and automatically identify required electronic components, their values, and relationship constraints.

FR-3: Knowledge Base Mapping

The system shall map identified entities to a predefined Component Library and apply predefined circuit-design and connection rules to determine logical wiring.

FR-4: Automatic Circuit Generation

The system shall automatically generate circuit connections based on the mapped logic and render the generated circuit in an understandable visual circuit-diagram format.

FR-5: Educational Output & Documentation

The system shall display detailed component and connection information and provide basic explanations of the generated circuit to help students understand its operation.

4. Non-Functional Requirements

4.1 Hardware Requirements

Processor: Intel Core i3 or above (Intel/AMD x64 processor).

RAM: Minimum 8 GB.

Storage: 256 GB SSD minimum.

Display: 14" or above.

Network: Internet connection required for downloading tools and optional online AI services.

4.2 Software & Technology Stack

Operating System: Windows 10/11 or Linux.

Frontend: HTML, CSS, JavaScript (with optional React.js framework).

Backend: Python (3.10 or above) using Flask or FastAPI.

AI/LLM: Ollama (for local AI support).

Visualization/Design Tools: CircuitikZ (for diagram generation), KiCad (for schematic design/validation), SVG/Canvas.

5. Database & Knowledge Base Schema

5.1 Component Library Table

ComponentID (Primary Key)

ComponentName (e.g., LED, Resistor, Push Button, Battery)

SymbolData (SVG path or CircuitikZ reference)

DefaultValue

PinConfiguration

5.2 Connection Rules Table

RuleID (Primary Key)

SourceComponent

TargetComponent

AllowedTopology (e.g., Series, Parallel)

DesignConstraint (e.g., requires current limiting resistor)

5.3 Prompt History Table

PromptID (Primary Key)

UserSessionID

RawText

ExtractedComponentsJSON

GeneratedSchematicPath

Timestamp

6. Use Cases

User (Student / Beginner)

Enter natural-language circuit prompt.

View generated circuit diagram/schematic.

Read auto-generated circuit explanation and component list.

System Backend

Process Natural Language Prompt via NLP.

Identify components and requirements from the Knowledge Base.

Execute circuit design logic and connection rules.

Generate SVG/Canvas visual output.

7. Agile Development Plan (Group 5 - 4 Members)

Sprint 1: Core Processing & Logic (Weeks 1-2)

Goal: Develop the NLP extraction engine, component knowledge base, and backend API structure.

Member 1 (Patare Anushree): Basic web interface (HTML/CSS/JS).

Member 2 (Ambhore Praful): Prompt Processing (NLP) using Python and LLM (Ollama).

Member 3 (Sontakke Sanket): Knowledge Base (Component Library and Connection Rules).

Member 4 (Kachare Sejal): Flask/FastAPI backend framework and routing.

Sprint 2: Generation, Visualization & Refinement (Weeks 3-4)

Goal: Implement automatic circuit routing, graphical rendering, and educational explanations.

Member 1: SVG/Canvas frontend rendering.

Member 2: Basic text explanations of synthesized circuits.

Member 3: Circuit Design Logic algorithm for automatic connections.

Member 4: CircuitikZ / KiCad rendering pipelines and end-to-end validation.

8. Product Backlog

ID	User Story	Priority	Module	Assigned
PB-01	As a user, I want a text input box so I can type my natural language circuit request.	High	UI / Frontend	Member 1
PB-02	As the system, I need to process text prompts to identify entities like 'LED' and 'Push Button'.	High	NLP / LLM	Member 2
PB-03	As the system, I need a library of predefined electronic components and their parameters.	High	Knowledge Base	Member 3
PB-04	As the system, I need to map extracted entities to connection rules to form a valid circuit.	High	Design Logic	Member 3
PB-05	As a user, I want to see a generated visual circuit diagram based on my prompt.	High	Circuit Generation	Member 4
PB-06	As the backend, I need APIs to connect the user interface to the Python logic.	High	Backend API	Member 4
PB-07	As a student, I want to read a basic explanation of how my generated circuit works.	Medium	Explanation Output	Member 2
PB-08	As a user, I want a list of components required for the generated circuit.	Medium	Output Dashboard	Member 1
PB-09	As a developer, I want to establish version control using Git & GitHub for the team.	High	Infrastructure	All